

Lecture 08

Cloud Management Mechanisms

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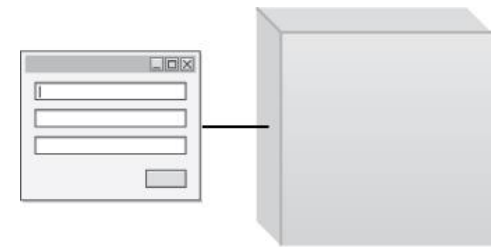
Cloud Management Mechanisms

- Mechanisms that encompass and enable cloud-based IT resources to be set-up, configured, maintained, and monitored.
- Facilitate the control and evolution of IT resources.
- Provide integrated API; offered as individual products, custom applications, or combined to product suites,
- Some of the mechanisms:
 - Remote Administration System
 - Resource Management System
 - SLA Management System
 - Billing Management System

Remote Administration System

Remote Administration System

- Provides **tools** and **user-interfaces** for external cloud resource administrators to configure and administer cloud-based IT resources.
- Can establish a portal for access to administration and management features of various underlying systems, including the resource management, SLA management, and billing management systems.
- Tools and APIs provided are generally used by the cloud provider to develop and customize online portals that provide cloud consumers with a variety of administrative controls.



remote administration
system

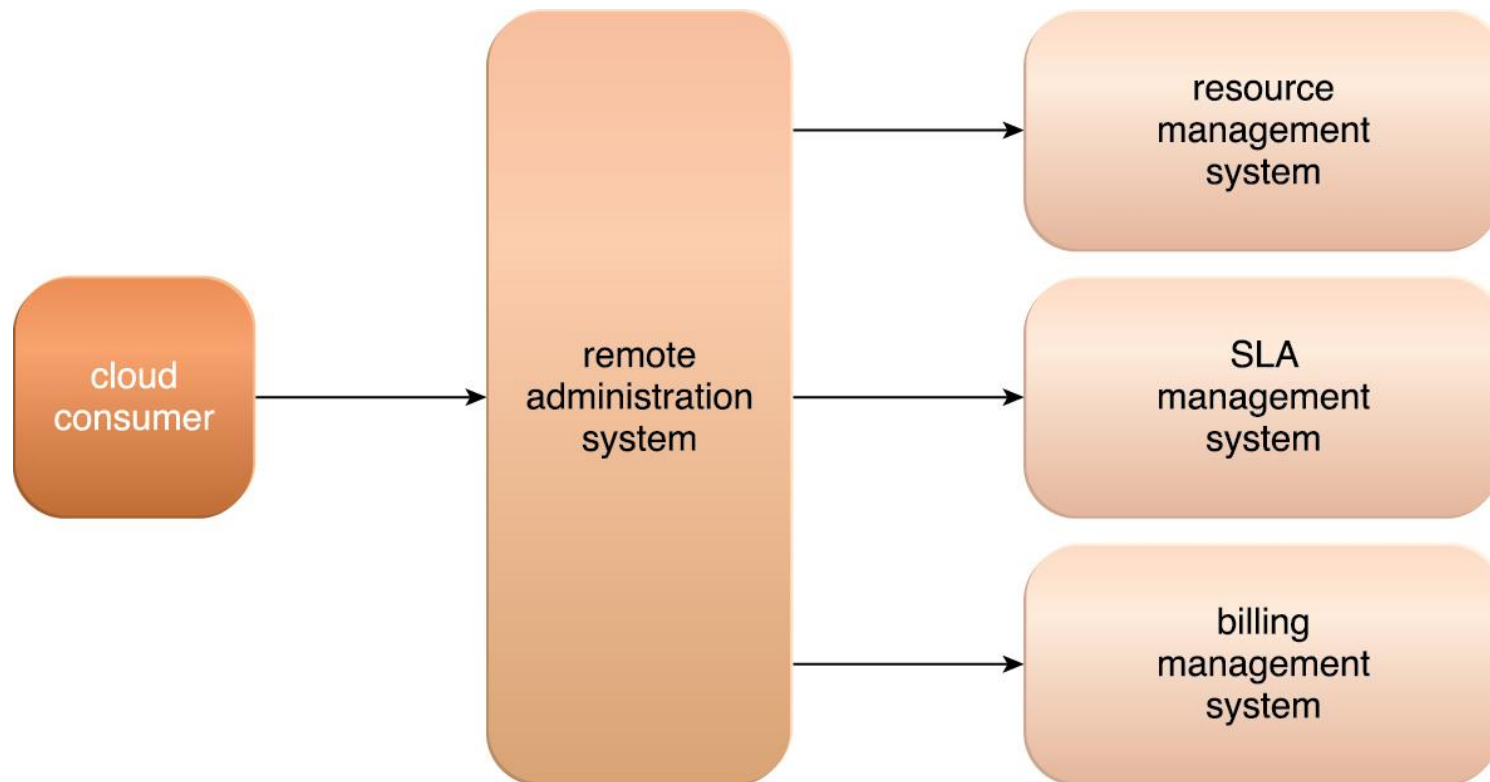


Figure 9.2 The remote administration system abstracts underlying management systems to expose and centralize administration controls to external cloud resource administrators. The system provides a customizable user console, while programmatically interfacing with underlying management systems via their APIs.

Portals

- Two types of portals that are created with the Remote Administration System:
 - **Usage and Administration Portal** – Centralizes management controls to different cloud-based IT resources and can further provide IT resource usage reports.
 - **Self-Service Portal** – A shopping portal that allows cloud consumers to search an up-to-date list of cloud services and IT resources that are available from a cloud provider (usually for lease), and submit their chosen items to the cloud provider for provisioning.

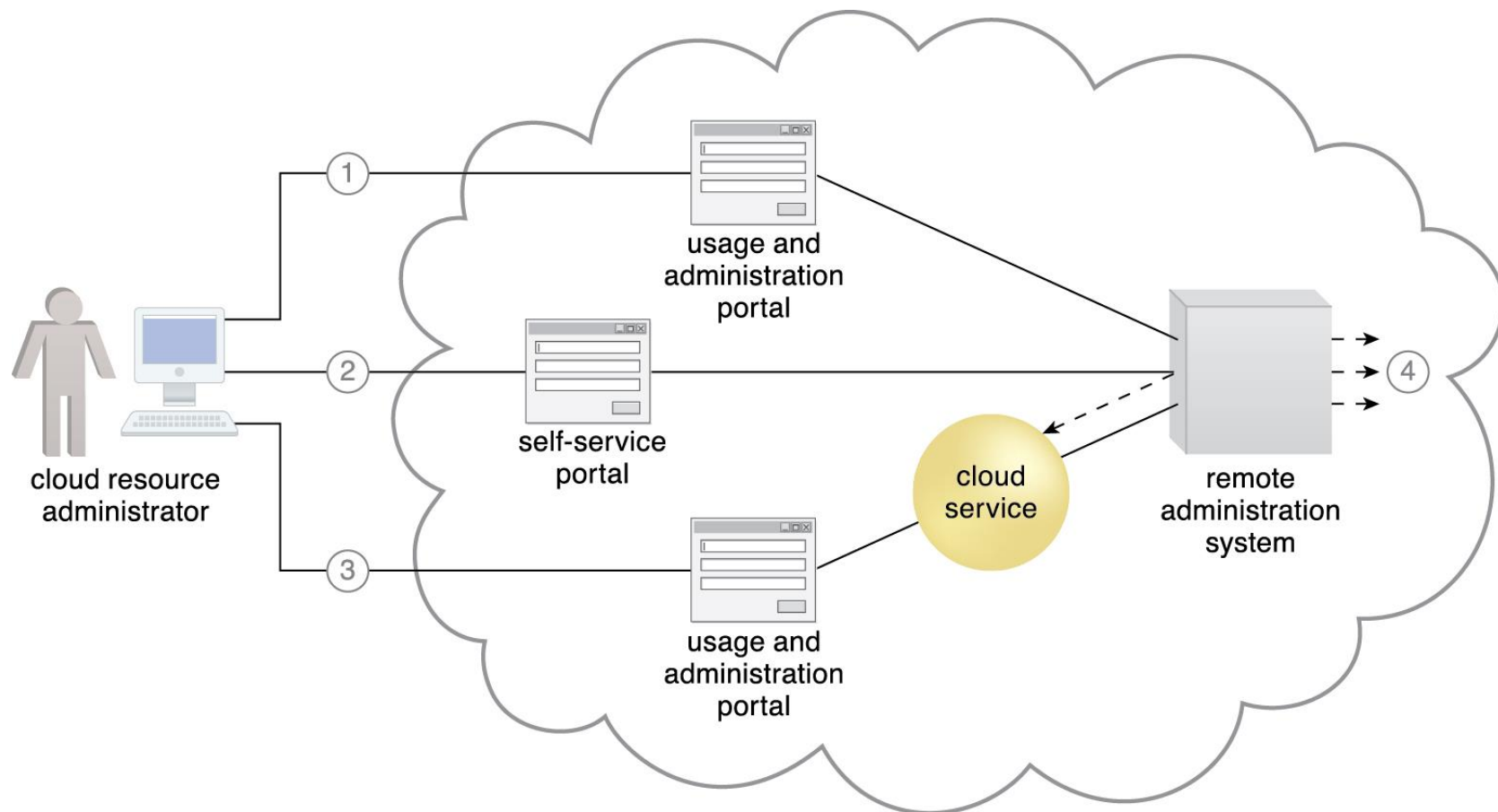


Figure 9.3 A cloud resource administrator uses the usage and administration portal to configure an already leased virtual server (not shown) to prepare it for hosting (1). The cloud resource administrator then uses the self-service portal to select and request the provisioning of a new cloud service (2). The cloud resource administrator then accesses the usage and administration portal again to configure the newly provisioned cloud service that is hosted on the virtual server (3). Throughout these steps, the remote administration system interacts with the necessary management systems to perform the requested actions (4).

- Tasks that can commonly be performed by cloud consumers via a remote administration control:
 - Configuring and setting up cloud services
 - Provisioning and releasing IT resource for on-demand cloud services
 - Monitoring cloud service status, usage, and performance
 - Monitoring QoS and SLA fulfilment
 - Managing leasing costs and usage fees
 - Managing user accounts, security credentials, authorization, and access control
 - Tracking internal and external access to leased services
 - Planning and assessing IT resource provisioning
 - Capacity planning

- Depending on:
 - The type of cloud product or cloud delivery model the cloud consumer is leasing or using from the cloud provider
 - The level of access control granted by the cloud provider to the cloud consumer.
 - Further depending on which underlying management systems the remote administration system interfaces with.

Standardized APIs

- The user interface by the remote administration system tend to be proprietary to the cloud provider.
- Some cloud consumers prefer to work with remote administration systems that offer standardized APIs (i.e. creation of its own front-end because:
 - Reuse the console if decides to move to another cloud provider that supports similar standardized API.
 - Leverage standardized APIs if leasing and centrally administer IT resources from multiple cloud providers and on-premise environments.

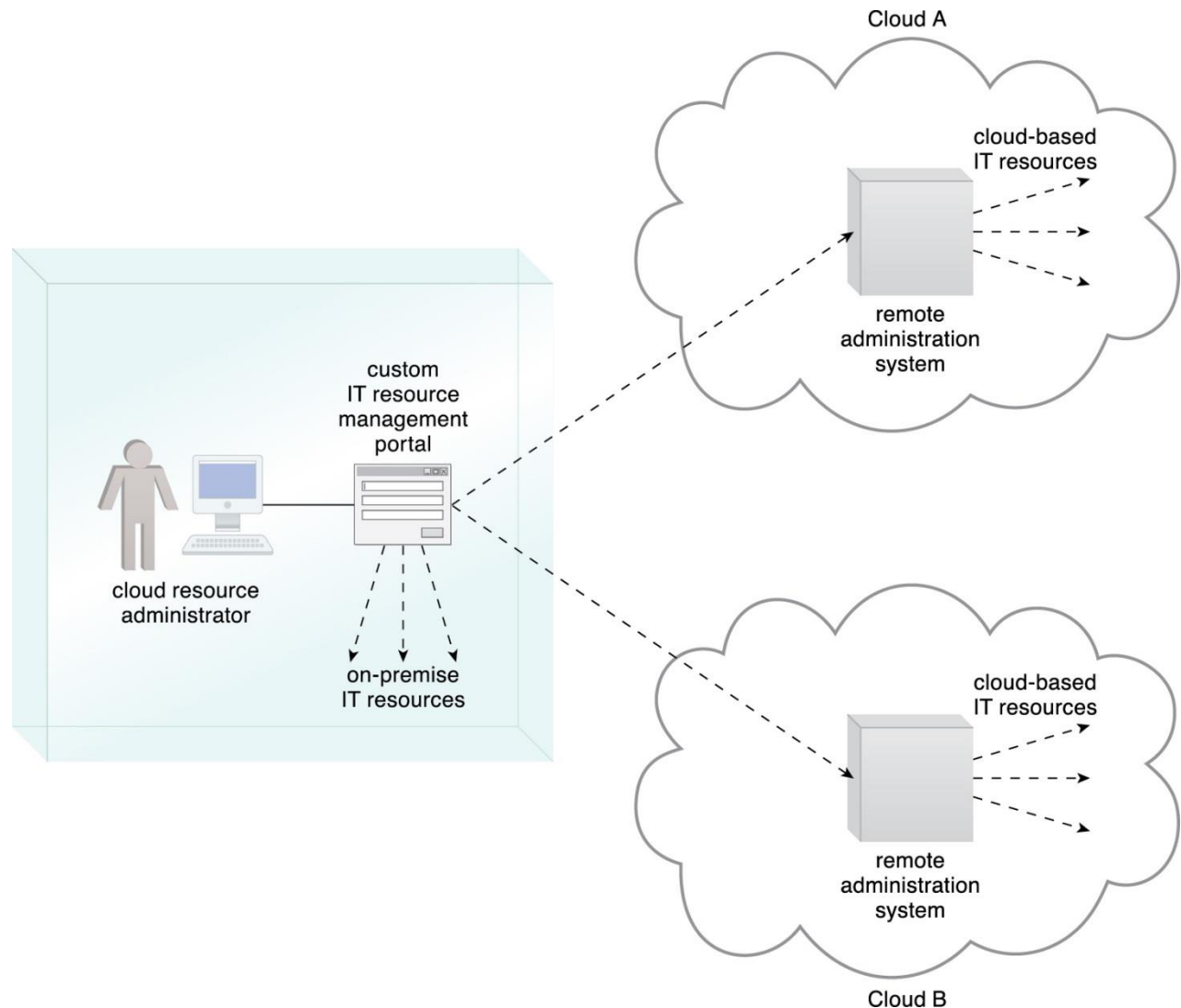


Figure 9.4 Standardized APIs published by remote administration systems from different clouds enable a cloud consumer to develop a custom portal that centralizes a single IT resource management portal for both cloud-based and on-premise IT resources.

Resource Management System

Resource Management System

- Coordinate IT resources in response to management actions performed by both cloud consumers and cloud providers.
- Core to this system is Virtual Infrastructure Manager (VIM) – a commercial product that can be used to manage a range of virtual IT resources across multiple physical servers.

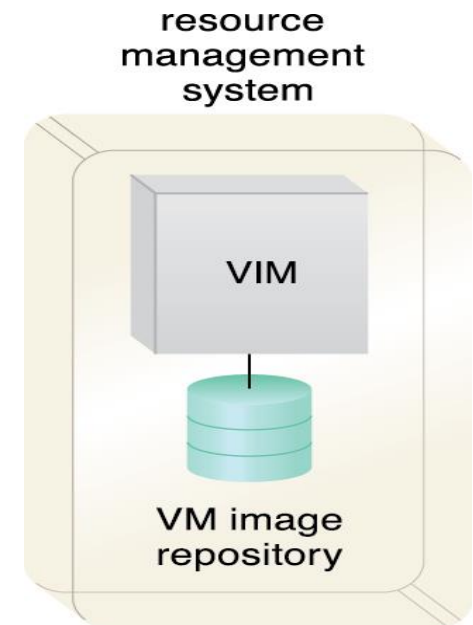


Figure 9.5 A resource management system encompassing a VIM platform and a virtual machine image repository. The VIM may have additional repositories, including one dedicated to storing operational data.

Tasks

- Managing virtual IT resource templates that are used to create pre-built instances, such as virtual server images.
- Allocating and releasing virtual IT resources into the available physical infrastructure in response to the starting, pausing, resuming, and termination of virtual IT resource instances.
- Coordinating IT resources in relation to the involvement of other mechanisms such as resource replication, load balancer, and failover system.
- Enforcing usage and security policies throughout the lifecycle of cloud service instances.
- Monitoring operational conditions of IT resources.

Resource Management System APIs

- Accessible by cloud resource administrators (RSAs) employed by cloud provider or cloud consumer.
- Resource management system's native console is accessible by the RSAs of cloud provider.
- Expose APIs that allow cloud providers to build remote administration system portals that are customizable to selectively offer resource management controls to external RSAs via usage and administration portals.

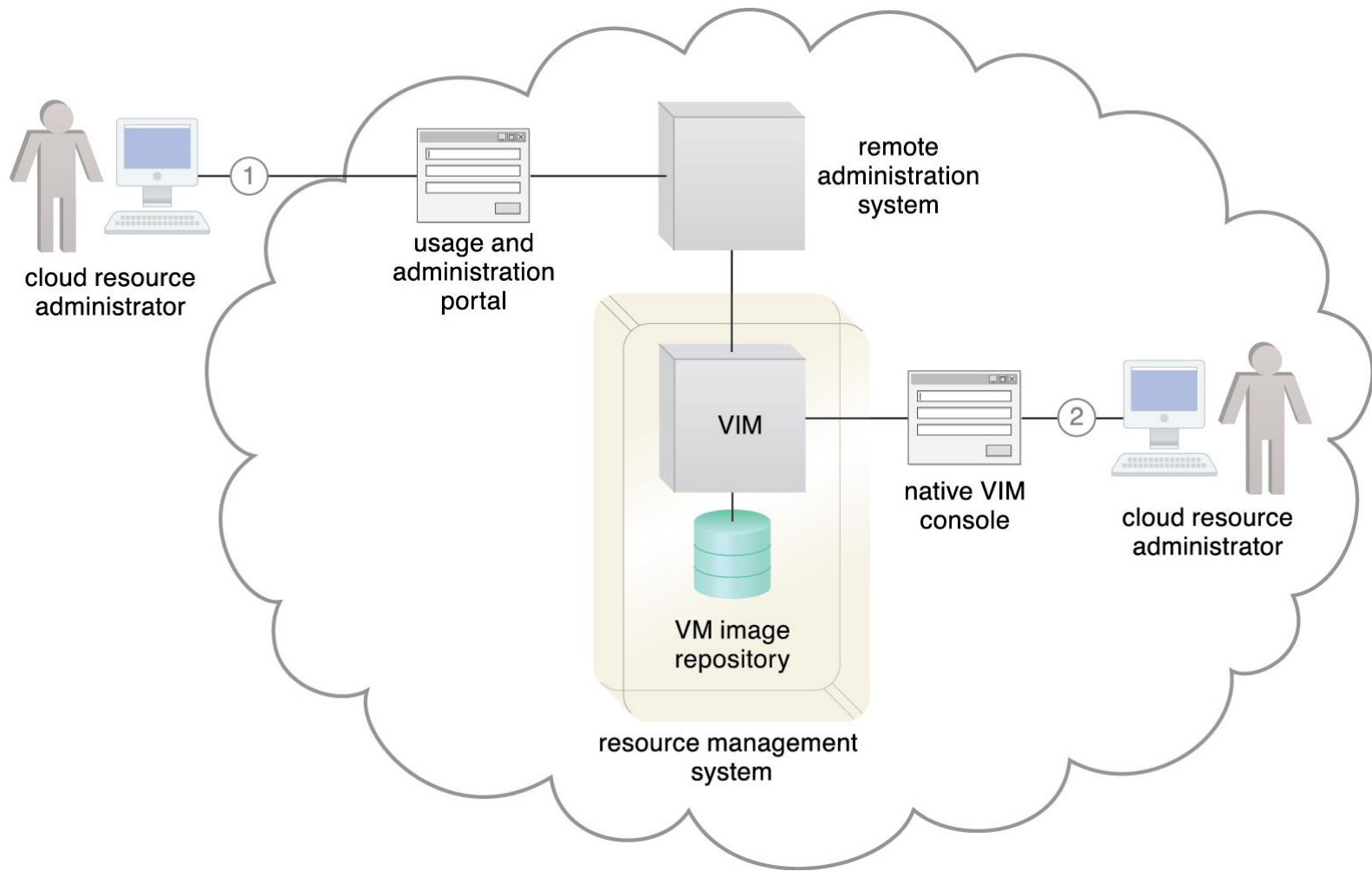


Figure 9.6 The cloud consumer's cloud resource administrator accesses a usage and administration portal externally to administer a leased IT resource (1). The cloud provider's cloud resource administrator uses the native user-interface provided by the VIM to perform internal resource management tasks (2).

SLA Management System

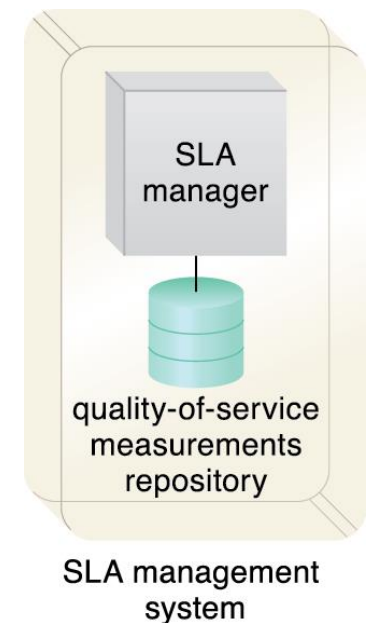
SLA Management System

- The **SLA management system** mechanism represents a range of commercially available cloud management products that provide features pertaining to the administration, collection, storage, reporting, and runtime notification of SLA data.
- Will generally include a repository used to store and retrieve collected SLA data based on pre-defined metrics and reporting parameters.

What is SLA?

A **service level agreement (SLA)** is a contract between a service provider (either internal or external) and the end user that defines the level of service expected from the service provider. **SLAs** are output-based in that their purpose is specifically to define what the customer will receive.

Figure 9.7 An SLA management system encompassing an SLA manager and QoS measurements repository.



SLA Monitor Mechanisms

- Rely on one or more SLA monitor mechanisms to collect the SLA data that can be made available in near-real time to usage and administration portals to provide on-going feedback regarding active cloud services.
- The metrics monitored for individual cloud services are aligned with the SLA guarantees in corresponding cloud provisioning contracts.

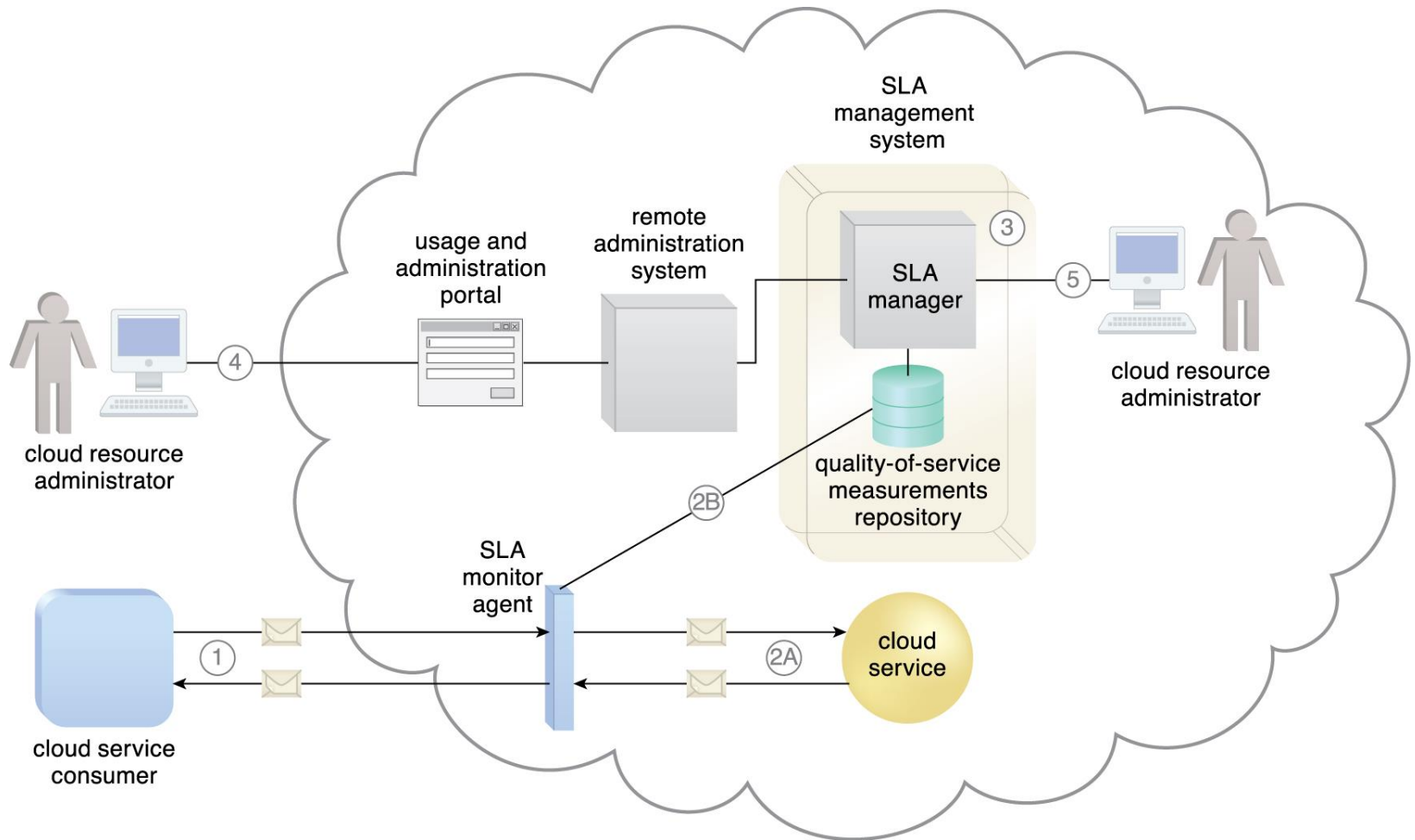


Figure 9.8 A cloud service consumer interacts with a cloud service (1). An SLA monitor intercepts the exchanged messages, evaluates the interaction, and collects relevant runtime data in relation to quality-of-service guarantees defined in the cloud service's SLA (2A). The data collected is stored in a repository (2B) that is part of the SLA management system (3). Queries can be issued and reports can be generated for an external cloud resource administrator via a usage and administration portal (4) or for an internal cloud resource administrator via the SLA management system's native user-interface (5).

Billing Management System

Billing Management System

- Dedicated to the collection and processing of usage data as it pertains to cloud provider accounting and cloud consumer billing.
- Relies on **pay-per-use monitors** to gather runtime usage data that is stored in a repository that the system use for billing, reporting, and invoicing purposes.
- Allows for the definition of different pricing policies, as well as custom pricing models on a per cloud consumer and / or per IT resource basis; using flat-rate or pay-per-allocation modes, or combination thereof.
- Billing arrangements be based on pre-usage and post-usage payments.
 - Post-usage can include pre-defined limits or unlimited usage (hence, unlimited billing).

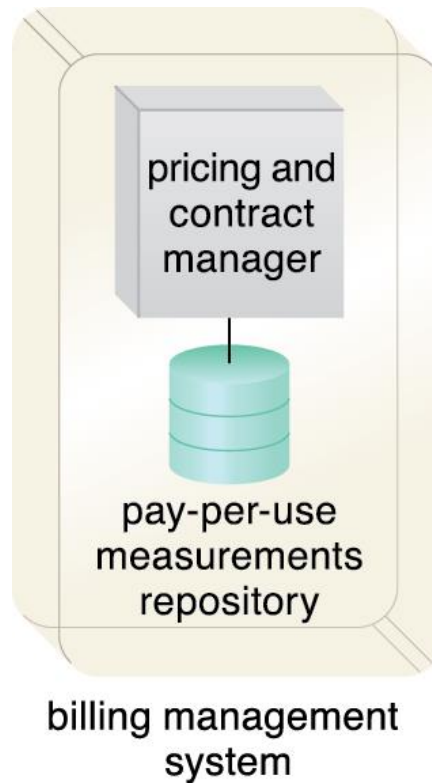


Figure 9.9 A billing management system comprised of a pricing and contract manager and a pay-per-use measurements repository.

Figure 9.10 A cloud service consumer exchanges messages with a cloud service (1). A pay-per-use monitor keeps track of the usage and collects data relevant to billing (2A), which is forwarded to a repository that is part of the billing management system (2B). The system periodically calculates the consolidated cloud service usage fees and generates an invoice for the cloud consumer (3). The invoice may be provided to the cloud consumer through the usage and administration portal (4).

